

Shivanshu Siyanwal

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 Uttar Pradesh, India

PROFESSIONAL SUMMARY

Quantum Software Engineer with 4+ years of experience in quantum algorithm development, production-grade Python software, and hybrid quantum-classical workflows. Proficient in Qiskit, Cirq, PennyLane, and cloud platforms (AWS Braket). Demonstrated expertise in quantum error correction, variational quantum algorithms (VQA/VQE/VQLS), and CI/CD for simulation infrastructure. Published researcher in QML, QKD, and photonic quantum computing. Seeking to develop and optimise scalable quantum software for real-world devices.

TECHNICAL SKILLS

Quantum Frameworks	Qiskit, Cirq, PennyLane, Perceval, Amazon Braket, CUDA-Q, QTensorFlow
Quantum Domains	Quantum error correction, VQAs, linear optical QC, quantum network simulation, tensor networks
Software Engineering	Python (5+ years, production code), C++, Git, CI/CD pipelines, Linux
ML/AI	PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn
Cloud & Backend	AWS Braket, FPGA deployment (AMD Alveo), HLS, HBM, REST APIs
Simulation Tools	OptiSystem, ANSYS, QuTip, QETLAB

WORK EXPERIENCE

CDAC (Centre for Development of Advanced Computing)

Noida, India

Quantum Applications Engineer

Oct 2024 – Present

- Developed hybrid quantum-classical pipelines for weather forecasting using VQC (7 qubits, 50 parameters), achieving 56.5% improvement over statistical baselines; production code deployed on cloud simulation infrastructure.
- Implemented and optimised quantum algorithms (VQE, QCNN, fidelity-based kernels) with Qiskit and PennyLane; built scalable software modules for quantum image encoding (FRQI, QPIE, NEQR).
- Deployed custom VQC kernel on AMD Alveo U55C FPGA using HLS and HBM, achieving 6.7× speedup over GPUs (291.8 s inference); integrated into CI/CD pipelines for continuous integration and testing.
- Collaborated with cross-functional teams (hardware, research, engineering) to optimise hybrid workflows and support cloud-based quantum computing platforms.

Dr. Arpita Maitra, IAI, TCG CREST

Kolkata, India

Project Junior Research Fellow – Quantum Networks & Linear Optics

Mar 2024 – Sep 2024

- Designed and simulated linear optical quantum network testbeds (DPS, COW, Twin-Field QKD) using OptiSystem and ANSYS, modelling realistic photonic channel impairments, loss, and noise.
- Performed error modelling and security analysis against intercept-resend and PNS attacks; contributed to resource state generation and entanglement generation benchmarking.
- Built Python post-processing tools to manage optical simulation data, enabling reusable and maintainable software for quantum network simulation.

Dr. Arpita Maitra, IAI, TCG CREST

Kolkata, India

Research Intern – Quantum Cryptography & Circuit Design

Aug 2023 – Mar 2024

- Designed quantum circuits for block ciphers (Mini-AES, S-AES) and Galois Field arithmetic; evaluated on Qiskit simulators and IBM Quantum hardware.
- Demonstrated chosen-ciphertext attack using Grover’s search and amplitude amplification; contributed to error correction and circuit optimisation.

Arqanum Technologies Pvt. Ltd.

Pune, India

Junior Quantum Computing Developer

Sep 2022 – Jul 2023

- Implemented and optimised VQC, QBM, and VQLS for time-series forecasting and financial analysis; used AWS Braket and Rigetti hardware for hybrid quantum-classical linear system solving.
- Developed quantum data encoding techniques (amplitude, phase, angle) and Ising Hamiltonian circuits; contributed to MeitY-QCAL project on Dicke state simulation.

RESEARCH EXPERIENCE

NMR QC Group, Prof. Kavita Dorai – IISER Mohali **Punjab, India**

MS Research Student – NMR Entanglement Detection & Quantum Simulation Aug 2021 – Aug 2022

- Designed neural network models for genuine multipartite entanglement detection and SLOCC classification in 2- and 3-qubit NMR systems; published in *Quantum Information Processing* (Springer).
- Performed experimental density matrix reconstruction and quantum error behaviour modelling; applied PCA, MCA, LDA for quantum dataset analysis.

EDUCATION

Indian Institute of Science Education and Research (IISER) Mohali **Punjab, India**

Five-Year Dual BS-MS (Postgraduate) – Major: Physics; Minor: Data Sciences Aug 2017 – May 2022

Dayawati Modi Academy (DMA-I)

Uttar Pradesh, India

High School Diploma (CBSE) – 95.8%

May 2014 – May 2017

SELECTED PUBLICATIONS & CONFERENCES

- **[IEEE TQCEBT 2026]** Hybrid Variational Quantum Circuits based Temperature Forecasting with Low-Latency FPGA Inference (accepted).
- **[IEEE AI-ML Bangalore 2025]** Quantum Machine Learning Models for Enhanced Weather Prediction.
- **[Springer Nature – Under Review]** Quantum Image Representations based Quantum Neural Networks for Binary Classification.
- **[Quantum Information Processing, Springer]** Entanglement Detection in 3-Qubit NMR States via ANNs (published).
- QTML 2024 (Univ. of Melbourne) – Poster: Quantum Techniques in Machine Learning.
- QCC2025 (IAI, TCG CREST) – Poster: R&D on Quantum Communication & Computation.

CERTIFICATIONS & AWARDS

- IBM Certified Associate Developer – Quantum Computation using Qiskit v0.2X (2026)
- Qiskit Global Summer School 2022 – Quantum Simulations (Advanced Level)
- Qiskit Global Summer School 2021 – Quantum Machine Learning (Certificate of Excellence)
- IBM Quantum Challenge Africa 2021 – Advanced Achievement
- INSPIRE Fellow (SHE) – DST, Ministry of Science & Technology, Govt of India (2017–2022)
- TOEFL iBT – Score 96 (2022)